

INSTITUTE OF ENGINEERING & MANAGEMENT,



KOLKATA



(A SCHOOL OF THE UNIVERSITY OF ENGINEERING AND MANAGEMENT, KOLKATA)

DETAILED SYLLABUS

1ST SEMESTER – MCA – 2026-2028 BATCH

3RD SEMESTER – MCA – 2025-2027 BATCH

DEPARTMENT OF COMPUTER APPLICATIONS

MCA STRUCTURED SYLLABUS - FIRST YEAR AND FIRST SEMESTER OF MCA2026 ADMISSION BATCH

Course Code	Course Title	Total No. of Weekly Contact Hours				Total No. of Credits
		Lecture (L)	Tutorial (T)	Practical (P)	Total Lectures	
1st Semester (Theory)						
MCA101	Computer Organisation and Architecture	4	1	0	5	3
MCA102	Computer Programming with C	4	1	0	5	4
MCA103	Data Structures with C	4	1	0	5	4
MCA105	Business English and Communication	3	1	0	4	3
MCA106	Computational Mathematics and Statistics	4	1	0	5	4
MCAGS101	Mental Maths for Professionals - I	2	1	0	3	1
Total of Theory					27	19
1st Semester (Practical)						
MCA192	C Programming Laboratory	0	0	4	4	3
MCA193	Data Structures Laboratory using C	0	0	3	3	3
Total of Practical					7	6
1st Semester (Sessional)						
MCA171	Research Methodology and IPR	2	1	0	3	2
MCA172	Python Programming Fundamentals	0	0	3	3	2
IFC	Industry and Foreign Certification	0	0	0	0	0
MAR	Mandatory Additional Requirements	0	0	0	0	0
MOOCs	Massive Open Online Course	0	0	0	0	0
Total of Sessional					6	4
Total of the Semester					40	29

DEPARTMENT OF COMPUTER APPLICATIONS

MCA STRUCTURED SYLLABUS - SECOND YEAR AND FIRST SEMESTER OF MCA2025 ADMISSION BATCH

Course Code	Course Title	Total No. of Contact Hours				Total No. of Credits
		Lecture (L)	Tutorial (T)	Practical (P)	Total Hours	
3 rd Semester (Theory)						
MCA301	Operating Systems and Systems Software	4	1	0	5	4
MCA302	Software Engineering & TQM	4	1	0	5	4
MCA304	Statistics and Numerical Techniques	3	1	0	4	3
MCA305	Artificial Intelligence and Machine Learning	4	1	0	5	4
MCAGS301	General Studies & Current Affairs-III	2	0	0	2	0.5
Total of Theory					21	15.5
3 rd Semester (Practical)						
MCA391	Operating Systems Laboratory (Unix)	0	0	2	2	3
MCA392	Software Project Management Laboratory	0	0	2	2	3
MCA395	Artificial Intelligence and Machine Learning Laboratory (PYTHON)	0	0	2	2	3
Total of Practical					6	9
3 rd Semester (Sessional)						
MCA371	Sustainability, Climate Action and Environmental Sciences	2	1	0	3	2
MCA381	Industrial Training	0	0	0	0	2
MCA382	Minor Project	0	0	0	8	6
MCA373	Seminar	0	0	0	0	1
MCAGS381	Competitive Aptitude Training - III	2	0	0	2	0.5
IFC	Industry and Foreign Certification	0	0	0	0	0
MAR	Mandatory Additional Requirements	0	0	0	0	0
MOOCS	Massive Open Online Courses	0	0	0	0	0
Total of Sessional					13	11.5
Total of the Semester					40	36

Subject Name: Computer Organisation and Architecture

Credit: 4

Subject Code: MCA101

Lecture Hours: 40

Course Name: Computer Organisation and Architecture	
Course Code: MCA101	Semester: 1st
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 4	

Aim:	
Sl. No.	
1	To have a thorough understanding of the basic structure and operation of a digital computer.
2	To study the different communication methods with I/O devices and standard I/O interfaces.
3	To learn the architecture and assembly language programming of 8085 microprocessor.

Objective:	
Sl. No.	
1	Understanding Logic gates, flip flops and counter.
2	Clear Understanding of Computer Architecture.
3	Clear Understanding of Pipeline processing, RISC and CISC architectures.
4	Develop a base for advanced microprocessors.
Pre-Requisite:	
Sl. No.	
1.	Proficiency in basic Digital Electronics
Course Outcome:	
1.	Summarize the fundamental components of a basic computer system and its organization.
2.	Apply arithmetic and logical microoperations of binary number systems.
3.	Analyze control unit design and concept of pipelining.
4.	Classify memory hierarchy and examine numerical problems based on it.

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-	-	-	-	2	-	2	-	-	-
CO2	3	2	2	2	-	-	1	-	-	2	-	2	-	-	-
CO3	2	2	3	3	-	-	1	-	-	2	-	2	-	-	-
CO4	3	2	3	2	-	-	-	-	-	2	-	2	-	-	-

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Structure of Computers and Computer Arithmetic	Computer types, Functional units, Basic operational concepts, von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point, Error detection and correction codes Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations.	International Academia: https://web.stanford.edu/dept/registrar/bulletin_past/bulletin02-03/pdf/CompSci.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4
2	Basic Computer Organization and Design	Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt. Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC	International Academia: https://web.stanford.edu/dept/registrar/bulletin_past/bulletin02-03/pdf/CompSci.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	8
3	Register Transfer, Micro-Operations and Micro-Programmed Control	Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit, Control Memory, Address Sequencing, Micro-Program example, Design of Control Unit.	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	8
4	Memory System:	Memory Hierarchy, Semiconductor Memories, RAM (Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Performance considerations, Virtual memory, Paging, Secondary Storage, RAID.	International Academia: https://web.stanford.edu/dept/registrar/bulletin_past/bulletin02-03/pdf/CompSci.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	7

5	Input-Output:	I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA.	International Academia: https://web.stanford.edu/dept/registrar/bulletin_past/bulletin02-03/pdf/CompSci.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	7
6	MULTI-PROCESSORS	Characteristics of multiprocessors, Interconnection structures, Inter Processor Arbitration, Interprocessor Communication and Synchronization, and Cache Coherence.	International Academia: https://web.stanford.edu/dept/registrar/bulletin_past/bulletin02-03/pdf/CompSci.pdf AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	6

List of Books Text Books:

Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
M. Moris Mano	Computer System Architecture	3 rd Ed	Pearson/PHI

Reference Books:

1. Carl Hamacher, Zvonks Vranesic, Safea Zaky (2002), Computer Organization, 5th edition, McGraw Hill, New Delhi, India.



Institute of Engineering & Management, Kolkata
Newtown
(School of University of Engineering and Management, Kolkata)



Subject Name: Computer Programming with C

Credit: 4

Subject Code: MCA102

Lecture Hours: 48

Course Name: Computer Programming with C & C Programming Laboratory	
Course Code: MCA102 & MCA192	Semester: 1st
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Practical: 4	Practical Sessional Internal continuous evaluation: 100
Credit: 4+3	Practical Sessional external examination: 100
Aim:	
Sl. No.	
1	To develop a rigorous understanding of algorithmic problem-solving and translate abstract logic into structured C programs.
2	To cultivate the ability to write modular, efficient, and memory-safe code suited for postgraduate-level software development.
3	To establish a strong foundation in core programming constructs, memory management, and data handling as a stepping stone for advanced systems programming.
Objective:	
Sl. No.	
1	A foundational understanding of computer operations, mathematical logic, and an aptitude for computational problem-solving.
2	To impart an in-depth understanding of memory management, specifically through pointer arithmetic and dynamic memory allocation.

3	To demonstrate the practical implementation of user-defined data types (structures and unions) and secure string manipulations.
4	To foster critical computational thinking necessary for optimising code, debugging errors, and implementing file handling for data persistence.
Pre-Requisite:	
Sl. No.	
1.	Proficiency in one high-level programming language
Course Outcome:	
Upon successful completion of this course, students will be able to:	
1.	Apply algorithmic thinking and fundamental C programming constructs (control structures, operators) to translate mathematical and logical problems into executable code.
2.	Analyse computational problems to select and implement appropriate linear data structures (arrays, strings) for efficient data organisation and manipulation.
3.	Evaluate memory management and program modularity by leveraging pointers, dynamic memory allocation, and user-defined functions to optimise software performance.
4.	Design and develop comprehensive software solutions utilizing heterogeneous data types (structures/unions) and persistent file I/O operations for real-world data processing scenarios.

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	2	1	-	1	1	1	1	2	3	1	2
CO2	2	3	3	2	2	1	-	1	1	1	1	2	3	2	2
CO3	2	2	3	2	3	1	1	1	1	1	2	2	3	2	2
CO4	2	2	3	2	3	2	1	1	2	1	2	2	3	3	2

Module Number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	Fundamentals & C Basics	Algorithmic Thinking & Compilation Process Tokens, Data Types, and Memory Footprints Formatted & Unformatted I/O Operators, Expressions, and Type Casting	International Academia: Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C) Harvard University: CS50 (Introduction to Computer Science) AICTE-prescribed syllabus (3 Years): AICTE-prescribed syllabus (2 Years): Industry Mapping: The concepts delivered are in sync with the industry standard	6	Week 1: Fundamentals, I/O, and Operators
2	Control Structures	if, else, and switch statements for, while, and do-while loops Nested loops and optimization Jump statements (break, continue, goto)	International Academia: Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C) Harvard University: CS50 (Introduction to Computer Science) AICTE-prescribed syllabus (3 Years): AICTE-prescribed syllabus (2 Years): Industry Mapping: The concepts delivered are in sync with the industry standards	6	Week 2: Control Structures
3	Functions & Storage Classes	Modular design, Declaration, and Call Stack Parameter passing and Return values Recursion mechanics and examples Storage Classes and Scope rules	International Academia: Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C) Harvard University: CS50 (Introduction to Computer Science) AICTE-prescribed syllabus (3 Years): AICTE-prescribed syllabus (2 Years): Industry Mapping: The concepts delivered are in sync with the industry standards	7	Week 3: Functions & Recursion
4	Arrays and Strings	1D Arrays and passing arrays to functions 2D Arrays and Matrix operations	International Academia: Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C) Harvard University: CS50	7	Week 4: 1-D and 2-D Arrays Week 5: Strings

		String basics and <string.h> library Secure coding (preventing buffer overflows)	<u>(Introduction to Computer Science)</u> <u>AICTE-prescribed syllabus (3 Years):</u> <u>AICTE-prescribed syllabus (2 Years):</u> Industry Mapping: The concepts delivered are in sync with the industry standards		& Secure Coding
5	Pointers & Dynamic Memory Allocation	Pointer fundamentals and Address mechanics Pointer arithmetic and advanced types (void, null) Pointers with Arrays and Strings Pass by Reference Dynamic Memory (malloc, calloc, free, leaks)	International Academia: <u>Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C)</u> <u>Harvard University: CS50 (Introduction to Computer Science)</u> <u>AICTE-prescribed syllabus (3 Years):</u> <u>AICTE-prescribed syllabus (2 Years):</u> Industry Mapping: The concepts delivered are in sync with the industry standards	9	Week 6: Pointers Fundamentals Week 7: Dynamic Memory Allocation (DMA)
6	User-Defined Data Types	Structures (Basics, Array of Structures, Nested) Passing Structures and the -> operator Memory alignment and padding Unions, enum, and typedef	International Academia: <u>Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C)</u> <u>Harvard University: CS50 (Introduction to Computer Science)</u> <u>AICTE-prescribed syllabus (3 Years):</u> <u>AICTE-prescribed syllabus (2 Years):</u> Industry Mapping: The concepts delivered are in sync with the industry standards	5	Week 8: User-Defined Data Types
7	File Handling & Advanced Concepts	File operations (Open, Read, Write, Close modes) Random Access (fseek, ftell, rewind) Command Line Arguments Preprocessor Directives	International Academia: <u>Massachusetts Institute of Technology (MIT): 6.087 (Practical Programming in C)</u> <u>Harvard University: CS50 (Introduction to Computer Science)</u> <u>AICTE-prescribed syllabus (3 Years):</u> <u>AICTE-prescribed syllabus (2 Years):</u> Industry Mapping: The concepts delivered are in sync with the industry standards	5	Week 9: File Handling & Command Line Arguments Weeks 10 & 11: Capstone Projects

List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
E. Balagurusamy	Programming in ANSI C	978-9389949186	McGraw-Hill Education
Yashavant Kanetkar	Let us C	978-9365899689	BPB Publications
K R Venugopal; Sudeep R Prasad	Mastering C	978-9332901278	McGraw-Hill Education
Reference Books:			
Dennis Ritchie; Brian W. Kernighan	C PROGRAMMING LANGUAGE, 2ND EDN	978-9332549449	Pearson Education India
K. N. King	C Programming, A Modern Approach, 2 nd Edition	978-0-393-97950-3	W. W. Norton & Company

Annexure – I

Course Code: MCA102

Course Name: C Programming Lab

Week 1: Fundamentals, I/O, and Operators

Direct Questions:

1. Write a program using the `sizeof()` operator to print the memory footprint of `char`, `int`, `float`, and `double` on your current compiler architecture.
2. Write a program to swap the values of two variables without using a third temporary variable.
3. Write a program that takes an integer input from the user and uses bitwise operators to determine if the number is odd or even.

Story-Based Question: A new music streaming startup needs a basic bandwidth calculator to estimate server load. Write a program that asks the user for the number of active listeners, the average audio bitrate (in kbps), and the average song duration (in minutes). Calculate and display the total data transferred in Megabytes (MB).

Week 2: Control Structures

Direct Questions:

1. Write a menu-driven program using a `switch-case` statement to simulate a basic calculator. Handle the divide-by-zero edge case.
2. Write a program to print a right-angled pyramid pattern of numbers.
3. Write a program to check if a user-inputted number is an Armstrong number.

Story-Based Question: A local research team is analyzing epidemiological data to track a recent outbreak. Write a program that takes a patient's data as input: Age, Body Temperature, and Symptom Duration. Use an `else-if` ladder to classify their risk level as "Low," "Moderate," or "High" based on threshold conditions you define.

Week 3: Functions & Recursion

Direct Questions:

1. Write a function `isPrime(int n)` that returns 1 if prime and 0 otherwise. Print all prime numbers between 1 and 100.
2. Write a recursive function to calculate the Factorial of a given number.
3. Write a user-defined function `power(int base, int exp)` that calculates base exponentiation without using `<math.h>`.

Story-Based Question: For an upcoming university CyberX workshop, participants need a secure, dynamically generated passcode. Write a recursive function that takes a 4-digit ID and recursively sums the digits until a single digit is obtained. Multiply that digit by a security constant to generate the passcode.

Week 4: 1-D and 2-D Arrays

Direct Questions:

1. Write a program to find the second largest and second smallest elements in a 1D array of integers.
2. Write a program to perform Matrix Multiplication on two 3x3 matrices.
3. Write a program to reverse the elements of a 1D array *in-place*.

Story-Based Question: You are organizing an online Capture the Flag (CTF) programming competition. You have two separate, sequentially sorted arrays of competitor scores. Write a program to seamlessly interleave and merge these two arrays into a single sorted array to generate the final leaderboard.

Week 5: Strings & Secure Coding

Direct Questions:

1. Write a program to find the length of a string and reverse it without using standard library functions.
2. Write a program that takes a string input and counts the total number of vowels, consonants, digits, and whitespaces.
3. Write a program to check if two strings inputted by the user are anagrams.

Story-Based Question: You are formatting an academic research paper on XAI, and the publisher requires specific proprietary frameworks to be redacted. Write a program that reads a text using `fgets()`, searches for a target word (like "SHAP" or "LIME"), and replaces all its occurrences with asterisks (****).

Week 6: Pointers Fundamentals

Direct Questions:

1. Write a program to swap two variables using pass-by-reference.
2. Traverse a 1D integer array and print its elements using *only* pointer arithmetic.
3. Write a function that returns both the maximum and minimum values of an array via pointer parameters.

Story-Based Question: A smart wearable device tracks `heart_rate` and `step_count`. Write a function taking these variables by reference. If the `heart_rate` exceeds 150 BPM, reset the `step_count` to zero and double the `heart_rate` to trigger a hardware alert state.

Week 7: Dynamic Memory Allocation (DMA)

Direct Questions:

1. Use `malloc()` to allocate memory for an array of N integers. Populate, calculate the sum, and `free()` the memory.
2. Demonstrate `calloc()` by allocating memory for an array of floats and verifying initialization to zero.
3. Use `realloc()` to expand a dynamically allocated array of 5 integers to hold 10 integers.

Story-Based Question: An epidemiological prognosis system needs to load daily patient records. Write a program that asks for the current day's patient count, dynamically allocates the memory to store their ages, calculates the moving average age, and strictly ensures the memory is freed before termination.

Week 8: User-Defined Data Types

Direct Questions:

1. Define a `struct Employee` (`emp_id`, `name`, `salary`). Read and print data for one employee.
2. Create an array of `struct Student`. Read data for 5 students and print the details of the one with the highest marks.
3. Write a program demonstrating a `union`, showing how memory is shared and overwritten.

Story-Based Question: An online music learning platform manages its catalog using structures. Define a `Course` structure containing an inner `Pricing` structure. Create an array of 3 courses, calculate the final price after discounts for each, and display the course with the lowest final cost.

Week 9: File Handling & Command Line Arguments

Direct Questions:

1. Write a program using command-line arguments to copy the contents of one file into another.
2. Write a program to open a text file and count the number of lines, words, and characters.
3. Accept an integer via command-line and append its multiplication table to `table.txt`.

Story-Based Question: A cybersecurity digital footprint analyzer monitors network requests. Write a program that reads a mock log file. Extract any line containing the keyword "UNAUTHORIZED" and append it to a new file called `threat_report.txt`.

Weeks 10 & 11: Capstone Projects

Students must select **ONE** of the following 12 capstone projects to complete. These projects require the integration of multiple core concepts including Pointers, Structures, Dynamic Memory, and File Handling.

Project 1: The Epidemiological Prognosis Tracker

An epidemiological research lab needs a system to process daily patient metrics. Write a C program that reads a dataset of patients (Patient ID, Age, Body Temperature, Symptom Duration) from a file. Using structures and pointer manipulation, apply a rules-based threshold to identify high-risk cases. The program must automatically extract these flagged records and generate a formatted "Prognosis_Report.txt" for the clinical team.

Project 2: Music Startup Ecosystem Manager

A new startup focusing on musical education and talent discovery needs a backend inventory engine. The platform manages three distinct entities: "Learning Courses", "Instrument Access", and "Talent Profiles". Using arrays of structures and dynamic memory allocation (`malloc/realloc`), build a menu-driven program that allows administrators to add new courses, update instrument availability, and display a sorted list of registered musical talents.

Project 3: CyberX Digital Footprint Analyzer

Following the successful conclusion of a CyberX cybersecurity workshop, the server generated a massive log file of network access requests. Write a command-line application that parses this "digital footprint" text file line by line. Using advanced string manipulation, the program must extract unique IP addresses, count their frequency, and quarantine any log entry containing the word "UNAUTHORIZED" into a separate secure text file.

Project 4: Capture The Flag (CTF) Auto-Scorer

You are tasked with building the backend scoring engine for an upcoming university Hackerrank-style CTF competition. Competitor data (Team Name, Challenges Solved, Time Penalties) is continuously updated. Use pointers to structures to manage the team data. Write a dynamic sorting algorithm that instantly updates the leaderboard based on net scores and displays the top 3 teams in real-time on the console.

Project 5: Explainable AI (XAI) Feature Importance Matrix

In machine learning, algorithms like SHAP and LIME output feature importance scores. Write a C program that simulates this by generating a 2D matrix of floating-point numbers representing "feature weights" for a dataset. Calculate the row-wise and column-wise averages. Apply a user-defined threshold to identify the most critical data features, extracting their indices into a 1D dynamically allocated array using pointers.

Project 6: University Event Poster Layout Generator

The Department of Computer Applications frequently hosts workshops and competitions. Write a text-based layout generator that takes user input for an event (Title, Date, Time, Venue, Speaker Name). Using string formatting, loop constructs, and standard I/O padding, the program must generate a visually appealing ASCII-art "poster" layout and save it as a text file to be distributed to the MCA students.

Project 7: Supermarket Inventory & Billing System

Build a robust point-of-sale system for a local supermarket. Define a structure for `Product` (ID, Name, Price, Stock). The program should allow a cashier to input purchased item IDs and quantities. It must check stock availability, calculate the total bill including a 5% GST, update the remaining inventory using pass-by-reference, and print an itemized receipt to the console.

Project 8: Advanced Cryptographic Secure Messaging

Design a two-way secure messaging utility. The program takes a string message from the user and a numeric encryption key. Implement a custom encryption algorithm that applies a Caesar Cipher shift followed by a bitwise XOR operation on every character. The program must be able to output the encrypted ciphertext, and conversely, take ciphertext and the key to decrypt it back to the original message.

Project 9: The Hospital Blood Bank Matcher

A regional hospital requires a fast blood-matching registry. Store donor records (Name, Blood Group, Age, Contact) in a binary file using `fwrite()`. When an emergency trauma request comes in, the user inputs the required blood type and units needed. The program must rapidly search the binary file using `fread()`, identify eligible donors, and output their contact information, ensuring all file pointers are securely closed.

Project 10: Dynamic Library Book Allocator

A university library's new acquisitions arrive in unpredictable batch sizes. Write a program that initially allocates memory for an array of 5 `Book` structures (Title, Author, ISBN). As the user inputs new books, if the array becomes full, use `realloc()` to dynamically double the array size without losing previous entries. Implement a function to safely `free()` the entire catalog when the program terminates.

Project 11: Railway Route & Seat Manager

Design a train ticketing system for a specific route. Define a structure for passenger tickets. Manage a 1D array representing 50 available seats. The menu-driven application must allow users to view available seats, book a specific seat number, cancel a ticket, and display the final passenger manifest. Prevent double-booking using logical condition checks.

Project 12: MCA Gradebook & SGPA Calculator

Create a comprehensive gradebook application for the 1st-semester MCA cohort. The program reads a file containing student Roll Numbers and marks in 5 subjects. Using user-defined functions, calculate the total marks, percentage, and SGPA for each student based on the university's grading scale. The application must generate individual text-file "report cards" named dynamically (e.g., "Report_Roll101.txt").



University of Engineering and Management

Institute of Engineering & Management, New Town Campus



Subject Name: Data Structure with C

Credit: 4

Code: MCA103

Lecture Hours: 40

Name of the Course: Data Structure with C & Data Structures Laboratory using C	
Course Code: MCA103 & MCA193	Semester: 1st
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Practical: 3	Practical Sessional Internal continuous evaluation: 100
Credit: 4+3	Practical Sessional external examination: 100

Aim:	
Sl. No.	
1	To gain Knowledge of Various aspects of algorithm development
2	To enhance Ability to identify qualities of a good solution
3	To implement learned algorithm design techniques and data structures to solve problems.
Objective:	
Sl. No.	
1	The fundamental design, analysis, and implementation of basic data structures.
2	Basic concepts in the specification and analysis of programs.

3	Principles for good program design, especially the uses of data abstraction.
4	Significance of algorithms in the computer field
Pre-Requisite:	
Sl. No.	
1.	Proficiency in one high level programming language
Course Outcome:	
1.	On completion of this course students are expected to learn various data structures, their usages, merits and limitations.
2.	On completion of this course students are expected to design and analyze various algorithms.
3.	On completion of this course students are expected to do a comparative analysis among different data structures and decide on the appropriate data structure to be used in a given scenario.
4.	On completion of this course students are expected to acquire adequate knowledge and skills to solve a real life software problem.

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	2	-	-	-	-	-	-	1	1	2	2
CO2	3	3	1	2	1	-	-	-	-	-	-	2	3	2	1
CO3	3	2	2	3	2	-	-	-	-	-	-	1	3	2	1
CO4	2	2	3	2	2	-	-	-	-	-	-	2	3	2	2

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	Algorithm Concept	Algorithm concept, Time Complexity, Space Complexity, Running Time– Worst Case, Best Case, Average Case, time space trade-off, Algorithm Efficiency- Linear loops, Logarithmic loops, Nested loops, Time complexity comparison- Polynomial vs Exponential, Algorithm Notations- Big O , Big Omega, Theta Notation	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makautexam.net/aicte_details/Syllabus/MCA/sem221.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4	
2	Introduction to Data Structure, Array	Program Efficiency, Data Structure-definition, usage, examples, Selection of Appropriate Data Structure, Data Structure-some terminologies, Classification of Data Structure, Fundamental difference between Linear and Non-linear Data Structure with examples, Operations on Linear Data Structure Introduction to Linear Data Structure-Array, 1D, 2D arrays, Row/Column major representation, sparse matrix	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makautexam.net/aicte_details/Syllabus/MCA/sem221.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	8	1. Write a C program to print an array. 2. Write a C program to check whether a given string is Palindrome or not. 3. Write a C program to convert temperature from degree Centigrade to Fahrenheit. 4. Write a C program to sort an array. 5. Write a C program to print the largest and second largest element of the array. 6. Write a C program to display Fibonacci series. 7. Write a program that reads two 2D metrics from the console, verifies if metrics multiplication is possible or not. Then multiplies the metrics and prints the 3rd metrics. 8. Write a program that reads a 2D metrics and checks if the metrics is a symmetric metrics or not. 9. Write a C program to print reverse array 10. Write a C program to check the sum of all elements of an array

					11. Write a C program to check duplicate number in an array. 12. Write a C program to read a 2D array (with most of the elements as 0s) and then represent the same array as Sparse Metrics. 13. Write a C program to pass an array to a function using Call by Value, update the array values in the function, print the array elements both in the function and in the calling function. 14. Write a C program to pass an array to a function using Call by Reference, update the array values in the function, print the array elements both in the function and in the calling function. 15. Write a program to display n number of elements. Memory should be allocated dynamically using malloc(). 16. Write a program to display n number of elements. Memory should be allocated dynamically using calloc(). 17. Write a program to allocate memory using malloc() and then reallocate the previously allocated memory using realloc(). Display the elements which have been taken after reallocation. 18. Write a program to allocate memory using calloc() and then reallocate the previously allocated memory using realloc(). Display the elements which have been taken after reallocation. 19. Write a C program to search an element in an Array using dynamic memory allocation
3	Linear Data Structure-Linked List	Linked List-Introduction, Representation, Memory Allocation, Types- Singly, circular, doubly, doubly & circular, Operations on various linked lists-Count, Traverse/Display, Search, Insert, Delete	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makautexam.net/aict	8	1. Write a Menu driven C program to accomplish the following functionalities in single linked list. a) Create a single linked list. b) Display the elements of a single linked list. c) Insert a node at the beginning of a single linked list.

[e_details/Syllabus/MCA/se
m221.pdf](#)

Industry Mapping: The concepts delivered are in sync with the industry standards

- d) Insert a node at the end of a single linked list.
 - e) Insert a node before a given node of a single linked list.
 - f) Insert a node after a given node of a single linked list.
 - g) Delete a node from the beginning of a single linked list.
 - h) Delete a node from the end of a single linked list.
 - i) Delete a node after a given node of a single linked list.
 - j) Delete the entire single linked list.
2. Write a Menu driven C program to accomplish the following functionalities in circular linked list.
- a) Create a circular linked list.
 - b) Display the elements of a circular linked list.
 - c) Insert a node at the beginning of a circular linked list.
 - d) Insert a node at the end of a circular linked list.
 - e) Delete a node from the beginning of a circular linked list.
 - f) Delete a node from the end of a circular linked list.
 - g) Delete a node after a given node of a circular linked list.
 - h) Delete the entire circular linked list.
3. Write a Menu driven C program to accomplish the following functionalities in doubly linked list.
- a) Create a doubly linked list.
 - b) Display the elements of a doubly linked list.
 - c) Insert a node at the beginning of a doubly linked list.
 - d) Insert a node at the end of a doubly linked list.
 - e) Insert a node before a given node of a doubly linked list.

					f) Insert a node after a given node of a doubly linked list. g) Delete a node from the beginning of a doubly linked list. h) Delete a node from the end of a doubly linked list. i) Delete a node after a given node of a doubly linked list. j) Delete the entire doubly linked list. 4. Write a Menu driven C program to accomplish the following functionalities in circular doubly linked list. a) Create a circular doubly linked list. b) Display the elements of a circular doubly linked list. c) Insert a node at the beginning of a circular doubly linked list. d) Insert a node at the end of a circular doubly linked list. e) Delete a node from the beginning of a circular doubly linked list. f) Delete a node from the end of a circular doubly linked list. g) Delete a node after a given node of a circular doubly linked list. h) Delete the entire circular doubly linked list.
4	Linear Data Structure-Stack	Introduction, Stack Operations – Push, Pop, Peek, Representation of Stack (Array, Linked List), Application of Stack: Reversing a list, Parentheses checker, Conversion of an infix expression into a postfix expression, Evaluation of a postfix expression, Conversion of an infix expression into a prefix Expression, Evaluation of a prefix expression, Recursion, Tower of Hanoi	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makauteexam.net/aicte_details/Syllabus/MCA/se_m221.pdf Industry Mapping: The concepts delivered are in	7	1. Write a Menu driven C program to accomplish the following functionalities in Stack using an Array: a. Insert an element into the stack using an array (Push Operation). b. Delete an element from the stack using an array (Pop Operation). c. Return the value of the topmost element of the stack (without deleting it from the stack) using an array. d. Display the elements of a stack using an array. 2. Write a Menu driven C program to accomplish the following functionalities in Stack using

			sync with the industry standards		Linked List: - Insert an element into the stack using a Linked List (Push Operation). - Delete an element from the stack using a Linked List (Pop Operation). - Return the value of the topmost element of the stack (without deleting it from the stack) using a Linked List. - Display the elements of the stack using a Linked List. - Write a program to convert an infix expression into its equivalent postfix notation. - Write a program to convert an infix expression into its equivalent prefix notation. - Write a program to evaluate a postfix expression. - Write a program to evaluate a prefix expression. - Write a program to print the Fibonacci series using recursion. - Write a program to solve the tower of Hanoi problem using recursion
5	Linear Data Structure-Queue	Introduction, Queue Operations – Enqueue, Dequeue, Peep, Representation of Queue (Array, Linked List), Types of Queues- Circular Queue, Deque, Priority Queue, Multiple Queue; Various operations (Enqueue, Dequeue, Peep) on the above mentioned queues-Both iterative & recursive implementation; Application of Queue	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makautexam.net/aicte_details/Syllabus/MCA/se m221.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	7	- Write a Menu driven C program to accomplish the following functionalities in Queue using an Array: - Insert an element into the queue using an array (Enqueue Operation). - Delete an element from the queue using an array (Dequeue Operation). - Return the value of the FRONT element of the queue (without deleting it from the queue) using an array (Peep operation) - Display the elements of a queue using an array. - Write a Menu driven C program to accomplish the following functionalities in Queue using Linked List: - Insert an element into the queue using a Linked List (Enqueue Operation). - Delete an element from the queue using a Linked List (Dequeue Operation). - Return the value of the FRONT element of

					the queue (without deleting it from the queue) using a Linked List (Peep operation). d. Display the elements of a queue using a Linked List. 3. Write a Menu driven C program to accomplish the following functionalities in Circular Queue using Array: - Insert an element into the circular queue. - Delete an element from the circular queue. - Return the value of the FRONT element of the circular queue (without deleting it from the queue). Display the elements of a circular queue using the circular queue
6	Searching & Sorting	Searching- Types of Searching (Linear Search, Binary Search, Interpolation Search), Comparison among various Searching techniques Sorting-Types, Methods (Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort), Technique, Explanation, Algorithm and Examples on various sorting methods, Comparison of various sorting algorithms in terms of time complexity (Average case, Worst case)	International Academia: https://web.stanford.edu/class/cs97si/03-data-structures.pdf AICTE-prescribed syllabus: https://makautexam.net/aicte_details/Syllabus/MCA/se m221.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	6	- Write a C program to implement the concept of Bubble sort. - Write a C program to implement the concept of Selection sort. - Write a C program to implement the concept of Insertion sort. - Write a C program to implement the concept of Quick sort. - Write a C program to implement the concept of Merge sort. - Write a C program to show that Quick sort is better than Bubble sort. - Write a C program to show that merge sort is more effective than quick sort. - Write a C program to search an element in an array using linear search. - Write a C program to search an element in an array using binary search. - Write a C program to search an element in an array using interpolation search.

List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
Reema Thareja	Data Structure Using C	2 nd Ed	Oxford
Reference Books:			
Tenenbaum	Data Structure Using C & C++	2 nd Ed	PEI
Kruse, Tondo & Leung	Data Structures & Program Design in C	2 nd Ed	PHI
Loudan	Mastering Algorithms With C		SPD/O'REILLY
Radhaganesan	C and Data Structures		Scitech Publications



University of Engineering and Management

Institute of Engineering & Management, New Town Campus



Subject Name: Business English and Communication

Credit: 3

Subject Code: MCA105

Lecture Hours: 33

Name of the Course: Business English and Communication	
Course Code: MCA105	Semester: 1
Duration: 33	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 3	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 3	

Aim:	
Sl. No.	
1	Making the students industry-ready.
2	Making the students relevant in the contemporary society.
3	Making the students prepared to analyze and solve problems through listening, speaking, reading and writing skills.
Objective:	
Sl. No.	
1	To develop effective business writing and communication skills.
2	To enhance oral communication and presentation abilities among students.
3	To help students learn to prepare various business documents and technical reports.
4	To improve listening and reading comprehension.
Pre-Requisite:	

Sl. No.	
1.	Basic English Proficiency, Listening and Speaking Skills, Reading and Writing Skills, Academic and Social Contexts, and Familiarity with Corporate Ethics.
Course Outcome:	
1.	Achieve competence in grammar, syntax, and vocabulary fundamentals.
2.	Effectively communicate in academic and social contexts.
3.	Develop readiness for the industry and understand corporate ethics.
4.	Acquire basic proficiency in English encompassing reading, listening, comprehension, writing, and speaking skills.

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	2	-	2	-	1	2	3	-	3	3	-	-
CO2	3	3	-	2	-	3	1	2	3	3	-	3	3	-	-
CO3	2	3	-	2	-	3	1	3	2	3	-	3	3	-	-
CO4	3	3	-	2	-	2	-	2	2	3	-	3	3	-	-

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Introduction to Business Communication.	- Importance of effective communication in business. - Types of business communication: Internal and External. - Communication process and barriers. - Strategies for effective communication. - Traditional and digital communication channels. - Effective use of email, memos, and business letters. - Communication through social media and professional networks.	International academia: https://www.coursera.org/learn/understanding-corporate-communications	4
2	Writing Skills Development	- Formats and styles of business letters. - Writing formal and informal business letters. - Common types of business letters: Inquiry, Complaint, Application, and Appreciation. - Structure of technical reports. - Writing abstracts, executive summaries, and conclusions. - Incorporating visuals and data in reports.	International Academia: https://ocw.mit.edu/courses/21g-222-expository-writing-for-bilingual-students-fall-2002/	8
3	Oral Communication Skills	- Preparing and delivering business presentations. - Using multimedia in presentations. - Techniques for effective public speaking. - Prepared speech exercises. - Extempore speech practice. - Role-playing business scenarios.	International Academia: https://ocw.mit.edu/courses/21g-222-expository-writing-for-bilingual-students-fall-2002/ Stanford Courses Online: https://online.stanford.edu/courses/csp-xcom88-high-impact-communication-advance-your-technology-career https://online.stanford.edu/courses/gsb-x0011-sharpen-your-communication-skills <i>Industry Mapping:</i> Campus Interviews and Recruitment Drives. Software: Orell Talk https://orelltalk.com/	8

4	Listening and Reading Skills	- Importance of active listening in business. - Techniques for improving listening skills. - Listening comprehension exercises. - Developing reading comprehension. - Strategies for effective reading. - Comprehension tests and exercises.	International Academia: https://ocw.mit.edu/courses/21g-222-expository-writing-for-bilingual-students-fall-2002/ <u>Stanford Courses Online:</u> https://online.stanford.edu/courses/csp-xcom88-high-impact-communication-advance-your-technology-career https://online.stanford.edu/courses/gsb-x0011-sharpen-your-communication-skills <i>Industry Mapping:</i> • Campus Interviews and recruitment drives. • Software: Orell Talk https://orelltalk.com/	
5	Practical Communication Applications	- Principles of organizing written material. - Structuring content for clarity and impact. - Editing and proofreading techniques - Designing effective posters for business presentations. - Visual and textual balance. - Presenting posters in professional settings.	<i>International Academia:</i> https://ocw.mit.edu/courses/21g-222-expository-writing-for-bilingual-students-fall-2002/ <i>Industry Mapping:</i> • Email writing and writing other relevant corporate documents. Software: Orell Talk https://orelltalk.com/	7

6	Practical Communication Skill Development	- Interactive sessions on negotiation and persuasion. - Group discussions and teamwork exercises	International academia: https://ocw.mit.edu/courses/15-280-communication-for-managers-fall-2016/ <i>Industry Mapping:</i> • Campus Interviews and recruitment drives.	6
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List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
R C Sharma and Krishna Mohan	Business Correspondence & Report Writing	ISBN 978-9385965050 (5 th ed)	McGraw Hill Education
Reference Books:			
Matthukutty Monippally	Business Communication Strategies	ISBN 978-0070435773	McGraw Hill Education
K.R. Lakshminarayanan	English for Technical Communication	Volume 1 & 2 Combined Edition	SCITECH PUBLICATIONS (INDIA) PVT LTD
Asha Kaul	Business Communication	Second Edition	PHI Learning
Dr. Anjali Ghanekar	Communication Skills for Effective Management	ISBN 978-8186314500 (19 th ed)	Everest Publishing House



University of Engineering and Management

Institute of Engineering & Management, New Town Campus



Subject Name: Computational Mathematics and Statistics

Credit: 4

Lecture Hours: 48

Subject Code: MCA106

Course Name: Computational Mathematics and Statistics	
Course Code: MCA106	Semester: 1 st
Duration: 48 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 4	

Aim:	
Sl. No.	
1	To provide a rigorous foundation in discrete mathematical structures, including set theory and propositional logic, which are essential for software development and formal methods.
2	To develop analytical and statistical problem-solving skills necessary for data-driven computing, focusing on probability distributions, parameter estimation, and statistical hypothesis testing.
3	To introduce advanced mathematical concepts like fuzzy sets and generating functions that underpin modern computational fields such as artificial intelligence, soft computing, and operations research.

Objective:	
Sl. No.	
1	Introduce the fundamental concepts of relations, functions, and propositional logic (including CNF and DNF) to formulate formal computational models and logic circuits.
2	Equip students with combinatorial tools, including the inclusion-exclusion principle, and mathematical induction techniques necessary for analyzing algorithm design and cryptography.
3	Teach foundational statistical measures, probability distributions (Binomial, Poisson, Normal), and hypothesis testing methods (z-test, t-test, chi-squared test, ANOVA) required for machine learning and numerical analysis.
4	Explore advanced mathematical frameworks, specifically generating functions for solving recurrences and fuzzy set operations for applications in financial modeling and soft computing.
Pre-Requisite:	
Sl. No.	
1.	A fundamental understanding of high-school-level algebra and introductory algorithmic thinking to effectively grasp abstract logic, recurrences, and numerical computations.
Course Outcome:	
Upon successful completion of this course, students will be able to:	
1.	Apply the principles of set theory, mathematical relations, and propositional logic to solve foundational discrete computational problems and model logic circuits.
2.	Analyze combinatorial problems and algorithm constraints using permutations, combinations, and the principle of mathematical induction.
3.	Evaluate datasets and statistical models by selecting and applying appropriate probability distributions and hypothesis testing methods (large and small sample tests) to assess data consistency and error.
4.	Formulate computational solutions for complex systems using advanced mathematical models, such as generating functions for recurrences and fuzzy sets for artificial intelligence applications.

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	-	-	1	2	-	1
CO2	3	3	2	1	-	-	-	-	-	-	-	1	2	-	1
CO3	3	3	1	3	2	-	-	-	-	-	-	1	1	3	2
CO4	3	3	2	2	1	-	-	-	-	-	-	2	1	2	3

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Set Theory and Functions	Set Theory: Definitions and operations (union, intersection, complement), Power sets and Cartesian products. Functions: Definitions and types (injective, surjective, bijective), Composition of functions and inverse functions.	Functions: Definitions and types (injective, surjective, bijective), Composition of functions and inverse functions.	4
2	Relations and Propositional Logic	Relations: Definitions and properties (reflexive, symmetric, transitive), Equivalence relations and partitions, Partial orders and Hasse diagrams. Propositional Logic: Propositions and logical connectives, Truth tables and logical equivalences, Normal forms (CNF, DNF).	Industry: Software development, logic circuits, artificial intelligence. Academia: Formal methods, logic in computer science	8
3	Combinatorics and Mathematical Inductions	Combinatorics: Permutations and combinations, Binomial theorem and Pascal's triangle, Inclusion-exclusion principle. Mathematical Inductions: Principle of mathematical induction,	Industry: Cryptography, algorithm design, network security. Academia: Discrete mathematics, theoretical computer science	8
4	Probability Distributions & Statistics	Statistics - measure of central tendency, dispersion (Moments, Skewness & Kurtosis). Least square curve fitting Introduction to mass function, density function, distribution function (Binomial, Poisson, Normal), estimation of parameters (unbiasedness-concept of noise/error, consistency). Testing of Hypothesis, Large Sample, Small Sample (z test, t test, chi-squared test, Anova)	Industry Mapping: https://www.sagemath.org/ , MATLAB International Academia: https://ocw.mit.edu/courses/18-440-probability-and-random-variables-spring-2014/pages/lecture-notes/ ,	6 6

			https://ocw.mit.edu/courses/2-993j-introduction-to-numerical-analysis-for-engineering-13-002j-spring-2005/pages/lecture-notes/	8
5	Advanced Topics	Generating Functions: Definitions and basic properties, Applications in counting and solving recurrences. Fuzzy Sets: Definitions and basic properties, Operations on fuzzy sets (union, intersection, complement), Applications and example.	Industry: Financial modelling, operations research. Academia: Discrete mathematics, combinatorial analysis, soft computing, artificial intelligence	6

University of Engineering and Management

Institute of Engineering & Management, New Town Campus

Subject Name: Mental Maths for Professionals

Credit: 1

Subject Code: MCAGS101

Lecture Hours: 48

Module number	Topic	Sub-topics	Mapping with International/National/ State Level Exams	Lecture Hours	Corresponding Assignment
1	Quantitative Aptitude	Textbook: Quantitative Aptitude, Author: R.S Aggarwal, Publisher: S.Chand A. Quant Foundation 1.Number System(Chapter 1) 2. HCF and LCM (Chapter 2) 3. Decimal Fractions (Chapter 3) 4. Simplification (Chapter 4) 5. Square roots and cube roots (Chapter 5) 6. Percentage (Chapter 11)- Basic concept of percentage & its shortcut rules & their applications. 7. Ratio and Proportion (Chapter 13)- Basic concept of Ratio & Proportion, Shortcut tricks & their	International Exams 1. GRE https://www.ets.org/gre/test-takers/general-test/prepare/content/verbal-reasoning.html#accordion-9f58105fc6-item-88093eca37) National Exams: 1. UPSC Civil Services Exam https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl-010223.pdf), pg 25-26 2. UPSC Combined Defence Services https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-21 3. Combined Graduate Level conducted by SSC https://ssc.nic.in/SSCFileServer/PortalManagement/Upload	24	1. Assignment on Numerical Problem Solving using Vedic Mathematics principle. 2. Assignment on Numerical Problem-Solving using percentage to fraction relation.

		applications. 8. Partnership (Chapter 14) concept, rules & Applications, Percentage Advanced problems & shortcuts. Profit & Loss (Chapter 12)- Basic concept, formulae, shortcut tricks & their application.	edFiles/notice CGLE 03042023.pdf) pg. 20-22 4. <i>Intelligence Bureau ACIO</i> (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) <i>State Level Exams:</i> 1. <i>Civil Services Executive Exam (WBCS)</i> (https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertise ment, pg 1 2. <i>Miscellaneous Services Recruitment Examination</i> (file:///C:/Users/UEMK/Downloads/2707970_2019.pdf) pg 1		
2	Logical Reasoning	Textbook: Modern Approach to Verbal and Non-Verbal Reasoning, Author Dr. R.S Aggarwal, Publisher: S.Chand 1. Coding and Decoding (Chapter 4) i. Conditional Coding, ii. Word-Pattern Coding, iii. Chinese Coding, 2. Direction Sense Test(Chapter 8) i. Direction Sense Test, ii. Direction Distance Test, iii. Shadow based Questions. 3. Series Completion (Chapter 1) i. Alphabet Series,	<i>International Exams</i> 1. <i>GRE</i> (https://www.ets.org/gre/test-takers/general-test/prepare/content/verbal-reasoning.html#accordion-9f58105fc6-item-88093eca37) <i>National Exams:</i> 1. <i>UPSC Civil Services Exam</i> (https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl-010223.pdf), pg 25-26 2. <i>UPSC Combined Defence Services</i> (https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-	24	1. Assignment on Letter Coding, Number Coding, Conditional Coding and Chinese Pattern. 2. Assignment on Directions and Distance 3. Assignment on Indicating based Blood Relation, Coding based Blood Relation and Family Tree based Blood Relation

		ii. Random Series, iii. Number Series, iv. Letter Gap, v. Missing Number Series, vi. Series Completion 4. Blood Relations (Chapter 5) – i. Family Tree Questions ii. Indication Type BR, iii. Coding Blood Relations, iv. Miscellaneous Blood Relations.	21 3. Combined Graduate Level conducted by SSC (https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_03042023.pdf) pg. 20-22 4. Intelligence Bureau ACIO (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) State Level Exams: 1. Civil Services Executive Exam (WBCS) (https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement, pg 1 Miscellaneous Services Recruitment Examination (file:///C:/Users/UEMK/Downloads/2707970_2019.pdf) pg 1		
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Learning Resources: Text Book

1. Quantitative Aptitude- R.S Agarwal
2. Verbal & non-verbal reasoning- R.S Agarwal



University of Engineering and Management
Institute of Engineering & Management, New Town Campus



Syllabus for MCA Admission Batch 2025, 1st Semester

Subject Name: RESEARCH METHODOLOGY AND IPR

Credit: 02

Subject Code: MCA171

Lecture Hours: 36 Hrs.

Name of the Course: RESEARCH METHODOLOGY AND IPR	
Course Code: MCA171	Semester: FIRST
Duration: 36	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 2	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 2	

Aim:	
Sl. No.	
1	To develop comprehensive understanding of research methodologies and systematic approaches to academic inquiry
2	To foster awareness of intellectual property rights and their application in research and development
3	To equip students with practical skills for conducting ethical and legally compliant research

Objective:	
Sl. No.	
1	To enable students to formulate well-defined research problems and design appropriate research frameworks
2	To provide knowledge of various data collection methods and their appropriate application in different research contexts
3	To develop competency in data analysis techniques and interpretation of research findings
4	To instil understanding of IPR principles, patent processes, and research ethics in academic and industrial settings
Pre-Requisite: Basic understanding of statistics and data interpretation. Familiarity with computer applications and software tools. Foundational knowledge of research concepts and academic writing	
Course Outcome:	
CO1	Formulate and design a research problem.
CO2	Understand and Comprehend the Data Collection Methods
CO3	Perform Data analysis and acquire Insights.
CO4	Understand IPR and follow research ethics.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	2	1	1	0	2	1	2	1	3	1	1	3
CO2	2	3	1	3	2	1	0	1	1	1	0	2	1	2	2
CO3	2	2	2	3	2	1	0	1	1	2	1	3	2	3	2
CO4	1	1	1	1	0	2	1	3	1	2	1	2	0	0	2

Module number	Topic	Sub-topics	Lecture Hours
1	Research Design	Overview of research process and design, Use of Secondary and exploratory data to answer the research question, Qualitative research, Observation studies, Experiments and Surveys.	6
2	Data Collection and Sources	Measurements, Measurement Scales, Questionnaires and Instruments, Sampling and methods. Data - Preparing, Exploring, examining and displaying.	6
3	Data Analysis and Reporting	Overview of Multivariate analysis, Hypotheses testing and Measures of Association. Presenting Insights and findings using written reports and oral presentations.	6
4	Intellectual Property Rights	Intellectual Property – The concept of IPR, Evolution and development of concept of IPR, IPR development process, Trade secrets, utility Models, IPR & Biodiversity, Role of WIPO and WTO in IPR establishments, Right of Property, Common rules of IPR practices, Types and Features of IPR Agreement, Trademark, Functions of UNESCO in IPR maintenance.	6
5	Patents	Patents – objectives and benefits of patent, Concept, features of patent, Inventive step, Specification, Types of patent application, process E-filing, Examination of patent, Grant of patent, Revocation, Equitable Assignments, Licences, Licensing of related patents, patent agents, Registration of patent agents.	6

List of Books Text Books:

Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
1. Cooper Donald R, Schindler Pamela S and Sharma JK, “Business Research Methods”, Tata McGraw Hill Education, 11e (2012).			

Reference Books:

1. Catherine J. Holland, “Intellectual property: Patents, Trademarks, Copyrights, Trade Secrets”, Entrepreneur Press, 2007.
2. David Hunt, Long Nguyen, Matthew Rodgers, “Patent searching: tools & techniques”, Wiley, 2007.
3. The Institute of Company Secretaries of India, Statutory body under an Act of parliament, “Professional Programme Intellectual Property Rights, Law and practice”, September 2013.

Syllabus for MCA Admission Batch 2026, 1st Semester

Subject Name: Basic Python Programming

Credit: 3

Subject Code: MCA172

Lecture Hours: 40

Name of the Course: Basic Python Programming	
Course Code: MCA172	Semester: 2nd
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Sessional: 2	End Semester Exam: 100
Practical: 3	Credit: 2

Aim:	
Sl. No.	
1	To provide a strong foundation in Python programming language and computational problem-solving.
2	To introduce procedural, modular, and object-oriented programming constructs available in Python.
3	To develop practical skills in building real-world applications using Python's standard library, file I/O, and database connectivity.

Objective:	
Sl. No.	
1	Understanding the basic syntax, variables, expressions, data types, and operators of Python.
2	Clear understanding of control flow constructs (conditionals, loops) and user-defined functions.
3	Proficient use of Python's core data structures — lists, tuples, dictionaries, and sets.
4	Understanding of modules, packages, file I/O, and exception handling in Python.
5	Develop a base for Object-Oriented Programming and modern application development using Python.

Pre-Requisite:	
Sl. No.	
1.	Basic knowledge of computer fundamentals.
2.	Familiarity with any introductory programming language (preferably C).
Course Outcome:	
1.	Understand Python fundamentals including data types, operators, and control-flow constructs.
2.	Develop modular Python programs using functions, scope rules, recursion, and standard library modules.
3.	Apply Python's core data structures (lists, tuples, dictionaries, sets) along with file I/O and exception handling to solve real-world problems.
4.	Design object-oriented solutions using classes, objects, inheritance, polymorphism, and constructors.

Relevant Links:

[Python for Everybody — Coursera \(University of Michigan\)](#)

[Programming, Data Structures and Algorithms using Python — NPTEL](#)

[Official Python 3 Documentation](#)

[Python Essential Training — LinkedIn Learning](#)

CO-PO-PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	3	-	-	-	-	2	-	2	3	-	-
CO2	3	3	2	2	3	-	-	-	-	2	-	2	3	2	-
CO3	3	3	3	3	3	-	-	-	-	2	-	2	3	2	-
CO4	3	3	3	3	3	-	-	-	-	2	-	2	3	3	2

Module-wise Syllabus Distribution

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Fundamentals & Basics	Introduction to Python: history, features, interpreter, interactive and script mode. Variables, expressions, statements, identifiers, keywords, comments. Standard data types: Numbers (int, float, complex), Booleans, Strings — basic string operations and slicing. Operators: arithmetic, relational, logical, assignment, bitwise, and membership operators. Input/output functions and type conversion.	International Academia: https://web.stanford.edu/class/cs41/ AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with industry standards for software development, data science, and automation.	7
2	Control Flow & Functions	Conditionals: if, else, elif, nested conditional statements. Loops: while and for loops, use of range(), break, continue,	International Academia: https://web.stanford.edu/class/cs41/ AICTE-prescribed syllabus:	8

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
		and pass statements. Functions: defining and calling functions, parameters and arguments (positional, keyword, default), return statement, local vs. global scope, lambda functions, recursion.	https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with industry standards for software development, data science, and automation.	
3	Core Data Structures (Collections)	Lists: creation, indexing, slicing, built-in methods, mutability, nested lists, list comprehension. Tuples: immutability, packing/unpacking, tuple methods. Dictionaries: key-value pairs, accessing and modifying elements, dictionary methods, dictionary comprehension. Sets: creation, set operations (union, intersection, difference), set methods.	International Academia: https://web.stanford.edu/class/cs41/ AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with industry standards for software development, data science, and automation.	9
4	Modules, Exception Handling & File I/O	Functions (advanced): keyword and default arguments, global/local scope, recursion (revisited). Modules: importing modules, from-import, packages, introduction to the Python Standard Library (math, random, os, datetime). Exception Handling: try/except/finally, handling common errors, raise statement, built-in exceptions, user-defined exceptions. File I/O: opening, reading, writing, and closing text and binary files; working with file pointers.	International Academia: https://web.stanford.edu/class/cs41/ AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with industry standards for software development, data science, and automation.	9
5	Advanced Programming (OOP)	Object-Oriented Programming in Python: classes and objects, attributes and methods, constructors (__init__), self parameter. Inheritance (single, multiple, multilevel), method overriding, polymorphism, encapsulation, abstraction. Introduction to modern tools: CRUD operations with MongoDB / building a simple web page with Django (project-level exposure).	International Academia: https://web.stanford.edu/class/cs41/ AICTE-prescribed syllabus: https://www.aicte-india.org/downloads/mcadegree.pdf Industry Mapping: The concepts delivered are in sync with industry standards for software development, data science, and automation.	7

List of Books			
Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
John V. Guttag	Introduction to Computation and Programming Using Python	3rd Edition	MIT Press / PHI
Sheetal Taneja	Python Programming: A Modular Approach	1st Edition	Pearson Education
Reference Books:			
Robert Sedgewick, Kevin Wayne, Robert Dondero	Introduction to Programming in Python: An Interdisciplinary Approach	1st Edition	Pearson / Addison-Wesley
Mark Lutz	Learning Python	5th Edition	O'Reilly Media
Allen B. Downey	Think Python: How to Think Like a Computer Scientist	2nd Edition	O'Reilly Media

Lab Assignments (MCA291)

Basics & Control Flow

1. Temperature Conversion: Write a program to convert Celsius to Fahrenheit and vice versa.
2. Prime Numbers: Check if a number is prime or print all primes in a given range.
3. Fibonacci Series: Generate the series up to n terms using loops or recursion.
4. Pattern Printing: Use nested loops to print star or number patterns.

Data Structures

5. List Operations: Create a list and perform operations like append, remove, sort, and find the largest/smallest element.
6. String Manipulation: Implement programs for vowel counting, string slicing, and palindrome checking.
7. Dictionary Application: Develop a student record system using names as keys and grades as values.

Advanced Concepts

8. Search & Sort: Implement Linear Search, Binary Search, and Selection or Bubble Sort.
9. OOP: Create a “Bank Account” or “Library Management” system using classes, objects, and inheritance.
10. File Handling: Read from and write to a text file (e.g., counting word frequency or reversing file contents).

Project / Modern Tools

11. Basic CRUD operations using MongoDB with Python.
12. Build a simple web page using the Django framework.

Story-based Assessment Questions (Unit 1 – Unit 5)

Unit 1: Fundamentals — The Online Store

Story: You are building a checkout system for a digital store. A customer buys an item with a price P and a tax rate T (percentage). Write a program to calculate the final price after adding tax and then applying a flat discount of ₹50.

Concepts: *Variables, Arithmetic Operators, Type Conversion.*

- Test Case 1: Price = 500, Tax = 10% → Expected Output: ₹500.0 (calculated as $500 + 50 - 50$)
- Test Case 2: Price = 100, Tax = 5% → Expected Output: ₹55.0 (calculated as $100 + 5 - 50$)

Unit 2: Control Flow & Functions — The Smart Fitness Tracker

Story: Your fitness tracker needs to give you a message based on your daily steps. Write a function `check_goal(steps)` that returns “Goal Reached” if steps are 10,000 or more, “Keep Walking” if between 5,000 and 9,999, and “Sedentary” if below 5,000.

Concepts: *Functions, if-elif-else, Comparison Operators.*

- Test Case 1: steps = 12000 → Expected Output: “Goal Reached”.
- Test Case 2: steps = 4500 → Expected Output: “Sedentary”.

Unit 3: Core Data Structures — The Movie Theater

Story: You have a list of available seats in a theater: `seats = [10, 12, 15, 20, 22]`. A user wants to book a specific seat number. Write a program that checks if the seat is available. If it is, remove it from the list and print “Booked”; otherwise, print “Unavailable”.

Concepts: *Lists, List Methods (remove, in operator).*

- Test Case 1: User Choice = 15 → Expected Output: “Booked”, Updated List: [10, 12, 20, 22].
- Test Case 2: User Choice = 11 → Expected Output: “Unavailable”.

Unit 4: Files & Exceptions — The Grade Reporter

Story: You are reading a file `grades.txt` that contains a list of numbers. Some lines might accidentally contain text instead of numbers. Write a program that reads the file, converts each line to an integer, and calculates the total sum. Use exception handling to skip any line that is not a valid number.

Concepts: *File I/O (open, readlines), try-except, ValueError.*

- Test Case 1: File contents `80\n90\nABC\n70` → Expected Output: Total Sum: 240 (skips ‘ABC’).
- Test Case 2: File is missing → Expected Output: “Error: File not found”.

Unit 5: Advanced Programming — The Virtual Bank

Story: Create a class BankAccount with an initial balance. It should have a method withdraw(amount) that reduces the balance but only if there are sufficient funds. If funds are insufficient, print “Insufficient Balance”.

Concepts: *OOP (Classes, Objects, Methods, self).*

- Test Case 1: Initial Balance = 1000, Withdraw = 400 → Expected Output: Balance: 600.
- Test Case 2: Initial Balance = 500, Withdraw = 600 → Expected Output: “Insufficient Balance”.



University of Engineering and Management
Institute of Engineering & Management, New Town Campus



Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: Operating System and System Software

Credit: 04

Subject Code: MCA301 & MCA391

Lecture Hours: 40 Hrs.

Name of the Course: Operating System and System Software & Operating Systems Laboratory (Unix)	
Course Code: MCA301 & MCA391	Semester: 3
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Practical: 2	Practical Sessional Internal continuous evaluation: 100
Credit: 4 +3	Practical Sessional external examination: 100

Aim:	
Sl. No.	
1	To understand the system architecture of an operating system
2	Ability to apply CPU scheduling algorithms to manage tasks.
3	Initiation into the process of applying memory management methods and allocation policies.
4	Knowledge of methods of prevention and recovery from a system deadlock.

Objective:	
Sl. No.	
1	To deliver a detailed knowledge of integral software in a computer system – Operating System.
2	To understand the workings of an operating system as a resource manager.
3	To familiarize the students with Process and Memory management.
4	To describe the problem of process synchronization and its solution.
Pre-Requisite:	
Sl. No.	
1.	You should know about Computer Architecture and Organization.
2	Proficiency in C or another programming language.
3	Familiarity with Assembly language.

Course Outcome:	
1.	Understand Operating System Concepts: Gain knowledge about operating system functions, generations, processes, and threads.
2.	Develop Process Scheduling Algorithms: Create algorithms for process scheduling, considering CPU utilization, throughput, turnaround time, waiting time, and response time.
3.	Identify the deadlock situation and provide an appropriate solution so that the protection and security of the operating system are also maintained.
4.	Learn File Handling and Process Control: Understand the basics of File, Device, and Disk Storage Management

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	2	1	0	0	1	1	0	3	2	0	1
CO2	3	3	3	2	2	1	0	0	1	1	0	2	3	1	1
CO3	3	3	2	2	2	2	1	2	1	1	0	2	3	1	1
CO4	3	2	3	1	2	1	0	0	1	1	0	3	3	0	1

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	Introduction	1. Introduction to Operating Systems 2. Hardware Support for Operating Systems 3. Resource Management 4. Operating System Architectures	International Academia: CS 372 Operating Systems Syllabus (utexas.edu); CS 140: Operating Systems (stanford.edu) AICTE-prescribed syllabus: mcadeegree.pdf (aict-india.org) Industry Mapping: The concepts delivered are in sync with the industry standards	4	• Basic Unix Commands
2	Process Management	5. Fundamentals of Process Management 6. Process Scheduling 7. Process Communication and Synchronization 8. Deadlocks 9. Multi-threading	International Academia: CS 372 Operating Systems Syllabus (utexas.edu); CS 140: Operating Systems (stanford.edu) AICTE-prescribed syllabus: mcadeegree.pdf (aict-india.org) Industry Mapping: The concepts delivered are in sync with the industry standards	8	• C Programs for Process Scheduling • Implementation of Banker's Algorithm
3	Memory Management	10. Basic Memory Management 11. Virtual Memory	International Academia: CS 372 Operating Systems Syllabus (utexas.edu); CS 140: Operating Systems (stanford.edu)	8	• C programs to simulate contiguous memory allocation techniques • C programs to simulate the paging technique

			AICTE-prescribed syllabus: mcadegree.pdf (aict-india.org) Industry Mapping: The concepts delivered are in sync with the industry standards		
4	File Management	12. File Systems 13. File System Implementation	International Academia: CS 372 Operating Systems Syllabus (utexas.edu); CS 140: Operating Systems (stanford.edu) AICTE-prescribed syllabus: mcadegree.pdf (aict-india.org) Industry Mapping: The concepts delivered are in sync with the industry standards	7	• Unix commands on file operations • C program for file organization technique.
5	Input –Output Management	14. Basics of I/O Management 15. Disk Management	International Academia: CS 372 Operating Systems Syllabus (utexas.edu); CS 140: Operating Systems (stanford.edu) AICTE-prescribed syllabus: mcadegree.pdf (aict-india.org) Industry Mapping: The concepts delivered are in sync with the industry standards	7	• C programs to simulate disk scheduling algorithms

6	Security and Protection Advanced Operating System	16. Security Issues 17. Protection Mechanisms 18. Distributed Operating Systems	<i>International Academia:</i> CS 372 Operating Systems Syllabus (utexas.edu) ; CS 140: Operating Systems (stanford.edu) <i>AICTE-prescribed syllabus:</i> mcadeegree.pdf (aicte-india.org) <i>Industry Mapping:</i> The concepts delivered are in sync with the industry standards	6	Advanced Unix commands
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List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
Naresh Chauhan	Principles of Operating Systems (Chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18)	1st Ed/ 9780198082873	Oxford University Press
Reference Books:			
Abraham Silberschatz, Peter B. Galvin	Operating System Concept	9th Ed/ 9788126554270	WILEY
Andrew S. Tanenbaum	Modern Operating Systems	4th Ed/ 9789332575776	Pearson Education India
William Stallings	Operating Systems	9th Ed/ 9789352866717	Pearson Education
Sumitabha Das	UNIX: Concepts and Applications (Lab Reference)	4th Ed/ 9780070635463	McGraw Hill Education



University of Engineering and Management

Institute of Engineering & Management, New Town Campus



Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: Software Engineering & TQM

Credit: 04

Subject Code: MCA302

Lecture Hours: 40 Hrs.

Name of the Course: Software Engineering & TQM	
Course Code: MCA302 & MCA392	Semester: 3rd
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Practical: 2	Practical Sessional internal continuous evaluation: 100
Credit: 4 + 3	Practical Sessional external examination: 100

Aim:	
Sl. No.	
1	To gain knowledge of various aspects of software engineering project management.
2	To enhance ability to identify qualities of a good solution
3	To implement learned algorithm/design techniques to solve problems

Objective:	
Sl. No.	
1	The fundamental knowledge of software engineering
2	The different basic models need to implement different project problems
3	The various design methods to develop the software system
4	The quality and other issues related to the software products and systems
Pre-Requisite:	
Sl. No.	
1.	Knowledge in fundamental theories of computer science and one programming language
Course Outcome:	
1.	On completion of this course students are expected to learn fundamentals and different models of software engineering.
2.	On completion of this course students are expected to learn different aspects of requirement analysis in software project management.
3.	On completion of this course students are expected to learn various types of software design and concepts of coding.
4.	On completion of this course students are expected to learn different types of testing and quality issues.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	1	-	2	1	3	-	-
CO2	2	3	2	1	-	-	-	1	2	3	3	1	3	-	-
CO3	3	2	3	-	2	-	-	-	-	-	-	1	3	-	-
CO4	2	2	2	3	2	-	-	1	1	1	-	1	3	-	-

Module Number	Topic	Subtopics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	Introduction and Software Process Models	Software, Software Engineering, Myths, Software Process, Work Products, Importance of Software Engineering, Standard for Software Process, Waterfall Model, Prototyping Model, Iterative Enhancement Model, Spiral Model, RAD model.	International Academia: (https://ocw.mit.edu/courses/16-355j-software-engineering-concepts-fall-2005/pages/lecture-notes/) AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/CS%20%28AI&ML%29.pdf Industry Mapping: IEEE SRS standard, Rational Rose, Reqview, Jira software, Axosoft.	8	1. Make a comparative studies of different models of software development process.
2	Requirement Engineering and Software Project Management	Software Requirements, Types of Requirements, Requirement Engineering Cycle, Requirements Specification document, Characteristics of Requirements, Requirement verification and validation, Role of Management in Software Development, Project Estimation Techniques, Staffing, Scheduling, Earned Value Analysis, Software Risks, Software Configuration Management, Software Process and Project metrics.	International Standards https://ocw.mit.edu/courses/16-355j-software-engineering-concepts-fall-2005/pages/lecture-notes/ AICTE prescribed syllabus: (https://www.aicte-india.org/sites/default/files/Model_Curriculum/CS%20%28AI&ML%29.pdf) Industry Mapping: MSproject, ProjectLibre, FunctionPointmodeler.	12	1. Write an SRS. 2. Compute function points using the method of FPA to determine the cost of s/w project 3. Implement COCOMO using the different formulas 4. Implement Gantt Chart and determine milestones 5. Implement PERT-CPM method

3	Software Design and Coding	Process, Data and Behavioural Modelling, Design Concepts, Modularity, Architectural design, Coupling and Cohesion, Top-down and bottom-up design, Object-oriented Analysis, Function-oriented and Object-Oriented Design approach, Software Design Document, Coding styles and documentation,	International Standards : (https://ocw.mit.edu/course/s/16-355j-software-engineering-concepts-fall-2005/pages/lecture-notes/) AICTE prescribed syllabus: (https://www.aicte-india.org/sites/default/files/Model_Curriculum/CS%20%28AI&ML%29.pdf) Industry Mapping: IEEE SDD document. Smart draw, Visual Paradigm/Microsoft Visio/MS Project/Umbrello/Rational Rose.	8	1. Implement the Cyclomatic Complexity of coding 2. Implement and evaluate the Halstead's Metrics of Coding 3. Implement Dharma's metrics 4. Implement polymorphism factor formula. 5. Implement inheritance formula
4	Testing and Software Quality	Testing principles, testing strategies, Black-box and White-box Testing Techniques, Levels of testing -unit, integration, system, regression, Test Plan, Test Cases Specification, Software debugging, Software Maintenance, Software Quality Factors, ISO , SEI CMM, CMMI, Software Reliability, Software Availability.	International Standards: (https://ocw.mit.edu/courses/6-170-laboratory-in-software-engineering-fall-2005/pages/assignments/) AICTE prescribed syllabus: (https://www.aicte-india.org/sites/default/files/Model_Curriculum/CS%20%28AI&ML%29.pdf) Industry Mapping: Eclipse, Bugzilla, MantisBT, Jira Software.	12	1. Implement H-K information factor. 2. Implement EMV method

List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
Rajib Mall	Fundamentals of Software Engineering(Chapter No. 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12)	4 th edn	PHI
Reference Books:			
Roger S. Pressman	Software Engineering, A Practitioners Approach(Chapter No. 8, 10, 14, 16, 26, 28)	7 th edn	MGH



University of Engineering and Management
Institute of Engineering & Management, New Town Campus
Syllabus for MCA Admission Batch 2025, 3rd Semester



Subject Name: Statistics and Numerical Techniques

Credit: 04

Subject Code: MCA304

Lecture Hours: 48 Hrs.

Name of the Course: Statistics and Numerical Techniques	
Course Code: MCA304	Semester: 3rd
Duration: One Semester	Maximum Marks: 100
Teaching Scheme: Lecture method	Examination Scheme
Theory: 4	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 4	
Aim:	
Sl. No.	
1	Equip students with the skills to collect, organize, and summarize data effectively, enabling them to understand the fundamentals of descriptive and inferential statistics.
2	Provide students with the knowledge of numerical techniques for solving complex mathematical problems, fostering proficiency in methods such as root finding, interpolation, and numerical integration.
3	Enable students to apply statistical and numerical methods to real-world scenarios across various disciplines, promoting critical thinking, problem-solving, and ethical data practices.

Objective:	
Sl. No.	
1	Develop students' understanding of different data types and the ability to collect, organize, and summarize data effectively, using descriptive statistics techniques.
2	Enable students to grasp the principles of statistical inference, including hypothesis testing, confidence intervals, and regression analysis, to draw meaningful conclusions from sample data about populations.
3	Equip students with proficiency in numerical techniques such as root finding, interpolation, and numerical integration, enabling them to solve complex mathematical problems encountered in various disciplines.
4	Foster the application of statistical and numerical methods in practical scenarios across diverse fields, through case studies and hands-on exercises, promoting critical thinking, problem-solving, and ethical data practices.
Pre-Requisite:	
Sl. No.	
1.	Basic knowledge of senior secondary and under graduate levels mathematics.
Course Outcome:	
1.	Upon completion of the course, students will demonstrate proficiency in collecting, analyzing, and interpreting data using appropriate statistical techniques, enhancing their ability to make informed decisions based on empirical evidence.
2.	Students will be able to apply numerical techniques like interpolation and numerical integration to solve complex mathematical problems encountered in engineering, science, and other disciplines, effectively utilizing computational tools to address real-world challenges.
3.	Students will be able to apply numerical techniques like solution of equation and system of linear equations to solve complex mathematical problems.
4.	At the end of the course, students will be able to apply numerical methods like numerical solution of ODE to solve complex mathematical problems encountered in engineering, science, and other disciplines to address day-to-day life critical problems.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	-	3	2	-	-	-	-	-	-	-			
CO2	3	2	-	2	3	-	-	-	-	-	-	-			
CO3	3	3	-	2	3	-	-	-	-	-	-	-			
CO4	3	3	-	2	3	-	-	-	-	-	-	-			

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Statistics, Probability and Distribution	Statistics - measure of central tendency, dispersion (Moments, Skewness & Kurtosis). Least square curve fitting - linear & non-linear. Probability, introduction to mass function, density function, distribution function (Binomial, Poisson, Normal), estimation of parameters (unbiasedness-concept of noise/error, consistency).	Industry Mapping: https://www.sagemath.org/ , MATLAB International Academia: https://ocw.mit.edu/courses/18-440-probability-and-random-variables-spring-2014/pages/lecture-notes/ , https://ocw.mit.edu/courses/2-993j-introduction-to-numerical-analysis-for-engineering-13-002j-spring-2005/pages/lecture-notes/	16
2	Interpolation and Numerical Integration	Interpolation-Newton's Forward, Backward, Sterling & Bessel's Interpolation formulae, Lagrange's Interpolation. Inverse Interpolation. Integration - Trapezoidal, Simpson's 1/3rd, Weddle's Rule, Romberg Integration, Gauss- Legendre two & three points formula, Newton Cotes Formula.	Industry Mapping: https://www.sagemath.org/ , MATLAB International Academia: https://ocw.mit.edu/courses/2-993j-introduction-to-numerical-analysis-for-engineering-13-002j-spring-2005/pages/lecture-notes/	12

3		Solution of any equation - Method of Iteration, Method of Bisection, Newton-Raphson Method, Regula-Falsi method and Secant Method. Solution of system of linear equations - Gauss Elimination Method, Gauss-Jacobi, Gauss-Seidel, LU factorization and Tri-diagonalization.	Industry Mapping: https://www.sagemath.org/, MATLAB International Academia: https://ocw.mit.edu/courses/2-993j-introduction-to-numerical-analysis-for-engineering-13-002j-spring-2005/pages/lecture-notes/	12
4		Solution of differential equations - Picard's method, Euler-modified method, Taylor's Series method, Runge-Kutta method, Milne's Predictor-Corrector method.	Industry Mapping: https://www.sagemath.org/, MATLAB International Academia: https://ocw.mit.edu/courses/2-993j-introduction-to-numerical-analysis-for-engineering-13-002j-spring-2005/pages/lecture-notes/	8

List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
B. S. Grewal	Higher Engineering Mathematics	44th Edition	Khanna Publishers
Reference Books:			
Dr. Hari Arora	PROBABILITY AND STATISTICS	3 rd Edition	S.K. KATARIA & SONS
K. DAS	NUMERICAL METHODS	2 nd Edition	U.N.DHUR & SONS PRIVATE LTD.
B.K. PAL & K. DAS	ENGINEERING MATHEMATICS Volume - IIA	1 st Edition (2021)	U.N.DHUR & SONS PRIVATE LTD.
Madhumangal Pal	Numerical Analysis for Scientists and Engineers: Theory and C Programs	1 st Edition (2007)	Alpha Science International Ltd

University of Engineering and Management

Institute of Engineering & Management, New Town Campus

Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: Artificial Intelligence and Machine Learning

Credit: 04

Subject Code: MCA305 & MCA395

Lecture Hours: 40 Hrs.

Name of the Course: Artificial Intelligence and Machine Learning	
Course Code: MCA305 & MCA395	Semester: 3rd
Duration: 40 Hrs.	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 4	End Semester Exam:100
Tutorial: 1	Continuous Assessment:100
Practical: 2	Practical Sessional internal continuous evaluation:100
Credit:4+3	Practical Sessional external examination:100

Aim:	
Sl. No.	
1	To gain knowledge of various aspects of artificial intelligence
2	To enhance the ability to identify qualities of a good solution of AI, ML etc.
3	To implement learned analytical techniques and AI & ML to solve problems.
Objective:	
Sl. No.	
1	Provide you with the knowledge and expertise to become a machine learning expert.
2	Demonstrate an understanding of intelligence and machine learning concepts that are vital for real-life problems.
3	Produce Python code to analyze different problems.
4	Critically evaluate intelligent problems based on their design, development and use in real-time.
Pre-Requisite:	
Sl. No.	
1.	Proficiency in Mathematics, Algorithms and Programming related to AI, ML etc.

Course Outcome:	
CO1.	Explain how knowledge is collected, created and managed for intelligence.
CO2.	Understand the key concepts in AI including their real-world applications and the toolkit used by intelligent systems.
CO3.	Implement AI & ML techniques using knowledge and toolkits.
CO4.	To enable systems to solve complex problems and automate decision-making processes.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	1	-	-	-	-	-	-	1	1	3	1
CO2	3	2	1	1	3	2	-	1	-	-	-	2	1	2	2
CO3	3	2	3	2	3	1	-	1	1	-	-	2	2	3	2
CO4	3	3	3	3	2	2	-	1	1	-	-	2	2	2	3

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignment
1	Module I: Introduction to AI(2L)	Overview of Artificial Intelligence – Introduction–History of AI– Applications of AI– Objectives of AI- features of AI	International Academia: https://ocw.mit.edu/course/res-str-002-data-management-spring-2016/ AICTE-prescribed syllabus: https://makauteexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in	4	1. Write a program for displaying reversal of a number. 2. Implement python script to read person's age from keyboard and display whether he is eligible for voting or not. 3. Implement python script to check the given year is leap year or not. 4. Implement Python Script to generate prime numbers series up to n. 5. To display elements of list in reverse order. 6. Implement python script to accept

			sync with the industry standards		line of text and find the number of characters, number of vowels and number of blank spaces in it. 7. Write a program which makes use of function to display all such numbers which are divisible by 7 but are not a multiple of 5, between 1000 and 2000. 8. Implement a python script for factorial of number by using recursion.
2	Module II: Symbolic Logic(6L)	Normal Forms in Propositional Logic – Logical Consequences – Resolution Principal – Predicate Calculus – Well Formed Formulas – Clausal Form – Rules of Inference – Unification – Resolution	International Academia: https://ocw.mit.edu/course/s/15-062-data-mining-spring-2003/ AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4	1. Write a program which accepts a sequence of comma-separated numbers from console and generate a list and a tuple which contains every number. Suppose the following input is supplied to the program: 34, 67, 55, 33, 12, 98. Then, the output should be: ['34', '67', '55', '33', '12', '98'] ('34', '67', '55', '33', '12', '98'). 2. Write Python script to copy file contents from one file to another. 3. Implement a python script to check the element is in the list or not by using Linear search & Binary search. 4. Implement a python script to arrange the elements in sorted order using Bubble, Selection, Insertion and Merge sorting techniques. 5. Write a python program by using exception handling mechanism. 6. Write a python program to perform various database operations (create, insert, delete, update).

3	Module III: Search Techniques (6L)	Different types of search techniques and comparison among them	International Academia: https://ocw.mit.edu/course/s/15-071-the-analytics-edge-spring-2017/pages/visualization/ AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	6	1. Write a program to compute summary statistics such as mean, median, mode, standard deviation and variance of the given different types of data. 2. Write a program to demonstrate Regression analysis with residual plots on a given data set.
4	Module IV: Knowledge Representation (8L)	Procedural versus declarative knowledge, forward versus backward reasoning, Structured Knowledge: Graphs, Frames, and Related Structures, Object-Oriented Representations, Representing knowledge in an uncertain domain, the semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics	International Academia: https://prolearn.mit.edu/data-science-and-big-data-analytics-making-data-driven-decisions AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4	1. Write a program to compute summary statistics such as mean, median, mode, standard deviation and variance of the given different types of data. 2. Write a program to demonstrate Regression analysis with residual plots on a given data set.
5	Module V: Expert systems(2L)	Characteristic features of expert systems Applications, importance of expert systems Rule based system architectures (the knowledge base, the inference process, explaining how or why, building a knowledge base, the I/O interface)	International Academia: AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in	4	

			sync with the industry standards		
6	Module VI: Machine Learning (10L)	Computational learning tasks for predictions, learning as function approximation, generalization concept. - Linear models and Nearest-Neighbors (learning algorithms and properties, regularization). - Neural Networks (MLP and deep models, SOM). - Probabilistic graphical models. - Principles of learning processes: elements of statistical learning theory, model validation. - Support Vector Machines and kernel-based models. To implement linear regression, Data classification, Data clustering – To learn how to create segments based on similarities using K-Means and Hierarchical clustering	International Academia: https://professionalprograms.mit.edu/online-program-internet-of-things/ AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4	1. Write a program to demonstrate the working of the decision tree-based ID3 algorithm. 2. Write a program to implement the Naïve Bayesian classifier for a sample training data set stored as a .CSV file. 3. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. 4. Write a program to implement k-Means clustering algorithm to cluster the set of data stored in .CSV file 5.
7.	Module VII: Introduction to Robotics(4L)	Definitions, illustration of application domains -Mechanics and kinematics of the robot- Sensors for robotics -Robot Control-Architectures for controlling behavior in robots- Robotic Navigation-Tactile Perception in humans and robots- Vision in humans and robots.	International Academia: https://ocw.mit.edu/course/s/6-034-artificial-intelligence-fall-2010/ AICTE-prescribed syllabus: https://makautexam.net/aicte_details/CourseStructure/MCA21.pdf Industry Mapping: The concepts delivered are in sync with the industry standards	4	Python lab for text analysis 1. Choose some book-length document and download it. 2. Count its characters, lines and words. 3. Count sentences, vocabulary, and the like. 4. Show collocations, common context, concordance, and similar relationships among the words. 5. Plot a lexical dispersion or two. 6. Plot a frequency distribution of the most common words.

List of Books Text Book			
Name of Author	Title of the Book	Edition	Name of the Publisher
Rich, Knight	Artificial Intelligence	3 rd edn	PHI
Patterson	Introduction to Artificial Intelligence and Expert Systems	2 nd edn	Pearson
Akerkar	Introduction to Artificial Intelligence	2 nd edn	PHI
Alpaydin	Introduction to Machine Learning	3 rd edn	PHI

Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: General Studies & Current Affairs - III

Credit: 0.5

Subject Code: MCAGS301

Lecture Hours: 48 Hrs.

Module number	Topic	Sub-topics	Mapping with International/ National/ State Level Exams	Lecture Hours	Corresponding Assignment
1	GK, Current Affairs and Economics	National income- Concept of GDP, GNP, NNP both in FC & MP, PCI Tax (BECC-103, Unit-1, Unit-2, Unit-3) http://egyankosh.ac.in/handle/123456789/67653 NCERT Textbook: (Chapter 2): https://ncert.nic.in/textbook.php?lecc1=2-6 Frank, ISC Economics (Chapter-14, 15, 16) https://books.google.co.in/books?id=4IGQISi9G7wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. Concept of tax, objective of tax, Direct & Indirect Tax, Progressive, Regressive & Proportional tax. 3. Textbook: Principles of Microeconomics: N Gregory Mankiw, Chapter 12) Textbook: FRANK Chapter-19 (class - 12) https://books.google.co.in/books?id=4IGQI	International Exams 1. GRE https://www.ets.org/pdfs/gre/gre-math-review.pdf) 2. GMAT https://downloads.mba.com/downloads/gmat-handbook.pdf) National Exams: 1. UPSC Civil Services Exam https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl-010223.pdf), pg 25-26 UPSC Combined Defence Services https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-21 3. RBI Grade B https://rbidocs.rbi.org.in/rdocs/Content/PDEs/DADVT/GRB09052023EA65E4FB1C2CE473396B4FD7E5F69	48	1. National Income: Write a report on the challenges and limitations in measuring national income, such as the informal sector, data collection issues, and non-market transactions. 2. Concept of Tax- Write a report on the difference between tax evasion and tax avoidance. Discuss the measures taken by the Indian government to combat tax evasion and promote tax compliance. 3. Inflation & Deflation Write a report on the causes of deflation and its consequences for the

	Si9G7wC&printsec=f rontcover&source=gb s_ge_summary_r&ca d=0#v=onepage&q&f=false 4. Inflation & Deflation - Inflation & its impact, Deflation & its impact, WPI, CPI, GDP deflator.(BECC-106, Block-2,Unit-6) http://egyankosh.ac.in/handle/123456789/75067 Textbook: M LJhingan 12th Edition. Macro-Economic Theory, Part-5,Chapter-37 Market structure-Perfect competition, monopoly,oligopoly, duopoly,monophony, duopoly, Oligopoly(BECC-101, Block-4, Unit-9,10,11,12) http://egyankosh.ac.in/handle/123456789/67491 Textbook: FRANKChapter-19 (Class-12) https://books.google.co.in/books?id=4lGQISi9G7wC&printsec=f rontcover&source=gb s_ge_summary_r&ca d=0#v=onepage&q&f=false GK and Current Affairs –Based on MonthlyMagazines provided andrecent news of national and international importance. Newspaper Reading: TheEconomic Times. Traditional GK and CA:Capitals of countries, the currency of countries, important dates, Sports football, hockey, recent events & awards etc. Important books &authors, ImportantHydropower dams, atomicpower plants, important national parks, Minister &portfolio &	CDDE.PDF), pg 22-23 4. IBPS Probationary officer(https://www.ibps.in/wp-content/uploads/Detailed-Advt.-CRP-PO-XII.pdf) ,Pg 7. 5. Combined Graduate Level conducted by SSC (https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_03042023.pdf) pg. 20-22 6. Intelligence Bureau ACIO (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) 7. XAT (https://xat.org.in/xat-syllabus/) 8. GATE (https://gate2025.iisc.ac.in/papers-and-syllabus/) 9. CAT https://iimcat.ac.in/per/g01/pub/756/ASM/WebPortal/1/index.html?756@@1@@1 State Level Exams: 1. Civil Services Executive Exam (WBCS) (https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement, pg 1	economy. 4. Market Structure. Analyze the impact of different market structures on consumers, focusing on factors like price, quality, and choice. Analyze their effectiveness and impact on the economy. ** All the assignments are in line with entrance exams for premier B-Schools and GS Paper-I of UPSC CSE.
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		constituencies,Population census, Persons in news -most famous,popular recent only, Important dances & festivals of Indian states,International Head Quarters & world organization, Importantpresident & pm electedfrom various countries, Important about banks likepayment banks, smallbanks & license system, Awards, Sports, Books & author, National & International affairs	2. Miscellaneous Services Recruitment Examination (file:///C:/Users/UEMK/Downloads/2707970_2019.pdf),pg 1		
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References

1. Indian Economy-Ramesh Singh



University of Engineering and Management
Institute of Engineering & Management, New Town Campus



Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: Sustainability, Climate Action and Environmental Sciences Credit: 03

Subject Code: MCA373

Lecture Hours: 36 Hrs.

Name of the Course: ENVIRONMENT AND ECOLOGY	
Course Code: MCA373	Semester: 3rd
Duration: 36	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 1	End Semester Exam: 100
Tutorial: 1	Continuous Assessment: 100
Credit: 2	
Aim:	
Sl. No.	
1	Imparting knowledge about the environment and ecosystem around us.
2	Imparting knowledge about the natural resources, biodiversity, and the importance of their conservation
3	Environmental Management and Pollution Control

Objective:	
Sl. No.	
1	Students will gain knowledge about the environment and ecosystem.
2	Students will learn about natural resources, biodiversity, and the importance of their conservation

3	To make students aware of problems of environmental pollution, its impact on humans and the ecosystem, and control measures.
4	At the end of the course, students will learn about waste disposal measures and environmental management.
Pre-Requisite: NA	
Course Outcome:	
1.	Define Environmental factors and the basic components of the ecosystem.
2.	Understand and explain the importance of Plantation.
3.	List the pollutants and analyze the importance of reducing/ controlling environmental pollution.
4.	Analyze the importance of Biohazards, Environmental and Social fety

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	3	2	1	2	1	2	2	2	3	2	1	2	2
CO2	3	2	3	2	2	3	1	2	2	1	1	1	3	1	3
CO3	2	2	1	3	1	2	3	3	1	1	2	3	3	3	1
CO4	1	3	1	3	3	2	2	3	2	3	2	1	1	1	2

Module number	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours
1	Overview	Basic ideas of environment, basic concepts, man, society & environment, their interrelationship Mathematics of population growth and associated problems, Importance of population study in environmental engineering, the definition of resource, types of resource, renewable, non-renewable, potentially renewable, effect of excessive use vis-à-vis population growth, Sustainable Development. Materials balance: Steady state conservation system, steady state system with non-conservative pollutants, step function. Importance, scope and principles of EIA.	International Academia: https://online.stanford.edu/courses/xeiet100-clean-renewable-energy-storage-sustainable-future AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: https://cbs.umn.edu/populus/downloadplant(WWTP) .	6

2	Ecology	Elements of ecology: System, open system, closed system, the definition of ecology, species, population, community, definition of ecosystem- components types and function. (1L) Structure and function of the following ecosystem: Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems, Mangrove ecosystem (special reference to Sundar ban); Food chain [definition and one example of each food chain], Food web.(2L) Biogeochemical Cycle- definition, significance, flow chart of different cycles with only elementary reaction [Oxygen, carbon, Nitrogen, Phosphate, Sulphur]. (1L) Biodiversity- types, importance, Endemic species, Biodiversity Hot-spot, Threats to biodiversity, Conservation of biodiversity.(2L)	International Academia: https://ocw.mit.edu/courses/1-020-ecology-ii-engineering-for-sustainability-spring-2008/ AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: https://vsni.co.uk/solutions/ecology https://www.helsinki.fi/en/researchgroups/statistical-ecology/software	6
3	Air Pollution	Atmospheric Composition: Troposphere, Stratosphere, Mesosphere, Thermosphere, Tropopause and Mesopause. (1L) Energy balance: Conductive and Convective heat transfer, radiation heat transfer, simple global temperature model [Earth as a black body, earth as albedo], Problems.(1L) Green house effects: Definition, impact of greenhouse gases on the global climate and consequently on sea water level, agriculture and marine food. Global warming and its consequence, Control of Global warming. Earth's heat budget.(1L) Lapse rate: Ambient lapse rate Adiabatic lapse rate, atmospheric stability, temperature inversion (radiation inversion).(2L) Atmospheric dispersion: Maximum mixing depth, ventilation coefficient, effective stack height, smokestack plumes and Gaussian plume model.(2L) Definition of pollutants and contaminants, Primary and secondary pollutants: emission standard, criteria pollutant. Sources and effect of different air pollutants- Suspended particulate matter, oxides of carbon, oxides of nitrogen, oxides of sulphur, particulate, PAN. (2L) Smog, Photochemical smog and London smog. Depletion Ozone layer: CFC, destruction of ozone layer by CFC, impact of other green-house gases, effect of ozone modification. (1L) Standards and control measures: Industrial, commercial and residential air quality standard, control measure (ESP. cyclone separator, bag house, catalytic converter, scrubber (ventury), Statement with brief reference). (1L)	International Academia: https://ocw.mit.edu/courses/1-84j-atmospheric-chemistry-fall-2013/pages/lecture-notes/ AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: https://www.who.int/europe/tools-and-toolkits/airq---software-tool-for-health-risk-assessment-of-air-pollution	6
4	Water Pollution	Pollutants of water, their origin and effects: Oxygen demanding	International Academia:	6

		wastes, pathogens, nutrients, Salts, thermal application, heavy metals, pesticides, volatile organic compounds. DO, 5-day BOD test, Seeded BOD test, BOD reaction rate constants, Effect of oxygen demanding wastes on river [deoxygenating, reaeration], COD, Oil, Greases, pH. Lake: Eutrophication [Definition, source and effect]. Waste water standard [BOD, COD], Water Treatment system, primary and secondary treatments, tertiary treatment definition. Water pollution due to the toxic elements. USEPA and WHO guidelines for drinking water.	https://online.stanford.edu/courses/cee270m-aquatic-and-organic-chemistry-environmental-engineering AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: Activated Sludge Simulation (ASIM), Sewage Treatment Operation and Analysis Over Time (STOAT), and GPS-X are the common softwares used for waste water treatment plant (WWTP).	
5	Lithosphere	Lithosphere; Internal structure of earth, rock and soil (1L). Solid Waste: Municipal, industrial, commercial, agricultural, domestic, pathological and hazardous solid wastes; Recovery and disposal method- Open dumping, Land filling, incineration, composting, recycling. Solid waste management and control (hazardous and biomedical waste). (2L)	International Academia: https://ocw.mit.edu/courses/1-34-waste-containment-and-remediation-technology-spring-2004/ AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: https://www.wasteworksonline.com/	6
6	Noise pollution	Definition of noise, effect of noise pollution, noise classification [Transport noise, occupational noise, neighbourhood noise] (1L) Definition of noise frequency, noise pressure, noise intensity, noise threshold limit value, equivalent noise level, L10 (18hr Index), n Ld. Noise pollution control. (1L)	International Academia: No link found AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: No software found	3
7	Environmental Management	Environmental impact assessment, Environmental Audit, Environmental laws and protection act of India, Different international environmental treaty/ agreement/ protocol. (2L)	International Academia: https://ocw.mit.edu/courses/11-601-introduction-to-environmental-policy-and-planning-fall-2016/ AICTE-prescribed syllabus: https://old.aicte-india.org/downloads/Environmental_Studies_curriculum.pdf Industry Mapping: https://www.intellex.com/products/environment/	3

List of Books Text Books:			
Name of Author	Title of the Book	Edition/ISSN/ISBN	Name of the Publisher
MP Poonia, SC Sharma, S. Kumar	Environmental Studies (AICTE Textbook)	3 rd - 2021/ 978-9390779024	Khanna Book Publishing Co.
Reference Books:			
NA			

University of Engineering and Management

Institute of Engineering & Management, New Town Campus

Syllabus for MCA Admission Batch 2025, 3rd Semester

Subject Name: Competitive Aptitude Training – III

Credit: 0.5

Subject Code: MCAGS381

Lecture Hours: 48 Hrs.

Module number	Topic	Sub- topics	Mapping with International/National/ State Level Exams	Lecture Hours	Corresponding Assignment
1	Quantitative Aptitude	Textbook: Quantitative Aptitude for Competitive Examination Author: R.S Agarwal Publishing House: S. Chand 1. Simple & Compound Interest, 2. Number System, 3. Quadratic Equations	National Exams: 1. <i>UPSC Civil Services Exam</i> (https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl-010223.pdf), pg 25-26 2. <i>UPSC Combined Defence Services</i> (https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-21 3. <i>Combined Graduate Level conducted by SSC</i> (https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_0304_2023.pdf) pg. 20-22 4. <i>Intelligence Bureau ACIO</i> (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) State Level Exams:	12	1. Simple & Compound Interest: Simple Interest Applications: - Calculate the total interest and amount payable on a loan with simple interest. - Determine the time required to double an investment with simple interest. - Compare the simple interest earned on different principal amounts or at different interest rates. Compound Interest Applications: - Calculate the compound interest and final amount of an investment over multiple years. - Compare the growth of an investment with different compounding frequencies (annual, semi-annual, quarterly, monthly). - Determine the time required to double or triple an investment with compound interest.
					Understand the concept of effective annual rate (EAR) and how it relates to nominal

			1. Civil Services Executive Exam (WBCS) https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement), pg 1 2. Miscellaneous Services Recruitment Examination https://adda247jobs-wp-assets-prod.adda247.com/jobs/wp-content/uploads/sites/7/2022/11/21142422/2707970_2019.pdf) pg 1		interest rates and compounding frequencies. 2. Number System: • Divisibility Rules: ○ Test the divisibility of numbers by 2, 3, 4, 5, 6, 8, 9, 10, and 11. ○ Apply divisibility rules to simplify calculations and solve problems. • Prime and Composite Numbers: ○ Identify prime and composite numbers. ○ Find the prime factorization of composite numbers. ○ Use prime factorization to find the highest common factor (HCF) and least common multiple (LCM) of numbers. • Number Properties: ○ Understand the concepts of even and odd numbers, natural numbers, whole numbers, integers, rational and irrational numbers. ○ Solve problems involving the properties of these number types. 3. Quadratic Equations: • Solving Quadratic Equations: ○ Solve quadratic equations using factoring, completing the square, and the quadratic formula. ○ Determine the nature of roots (real, equal, imaginary) of a quadratic equation. • Word Problems: ○ Apply quadratic equations to solve real-world problems, such as finding the dimensions of a rectangle given its area and
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					perimeter, or determining the trajectory of a projectile. 4. Quadratic Functions and Graphs: - Graph quadratic functions and interpret the graph to find the vertex, axis of symmetry, and intercepts. - Use the graph to solve quadratic equations and inequalities.
2	Logical Reasoning	Textbook: Verbal and Non-Verbal reasoning Author: R.S Agarwal Publishing House: S. Chand - Puzzle - Classification Based Puzzle - Sequential Based Puzzle - Selection Based Puzzle - Ranking Based Puzzle - Blood Relation Based Puzzle Inequality	National Exams: UPSC Civil Services Exam (https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl-010223.pdf), pg 25-26 UPSC Combined Defence Services (https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-21 Combined Graduate Level conducted by SSC (https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_0304_2023.pdf) pg. 20-22 Intelligence Bureau ACIO (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) State Level Exams: Civil Services Executive Exam (WBCS) (https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement), pg 1	12	1. Classification Based Puzzles: Grouping by Attributes: Provide a list of items (e.g., animals, fruits, professions) and ask students to classify them into groups based on shared characteristics (e.g., habitat, color, skill set). Identifying the Odd One Out: Present a group of items where one does not belong and have students explain why it is different from the others. Missing Item: Give a set of items with a pattern and have students determine the missing item that fits the pattern. 2. Sequential Based Puzzles: Logical Sequencing: Present a series of events or actions and have students arrange them in a logical order. Number Series: Give a series of numbers with a pattern and ask students to find the missing number or continue the series. Letter Series: Provide a series of letters with a pattern and have students determine the missing letter or continue the series. 3. Selection Based Puzzles: Team Selection: Provide a set of candidates with different skills and have students select the best team for a specific task. Item Selection: Give a list of items with

			2. Miscellaneous Services Recruitment Examination (file:///C:/Users/UEMK/Downloads/2707970_2019.pdf) <i>pg1</i>		different attributes and ask students to choose the most suitable item for a given purpose. • Eligibility Criteria: Present a set of rules or conditions and have students determine which candidates are eligible or ineligible based on those criteria. 4. Ranking Based Puzzles: • Height/Weight Arrangement: Arrange a group of people in ascending or descending order based on their height or weight. • Marks/Scores: Order students or players based on their marks, scores, or performance in a competition. • Preferences: Determine the order of preference for a group of people based on their likes and dislikes. 5. Blood Relation Based Puzzles: • Family Tree: Present a family tree with missing information and have students deduce the relationships between different members. • Coded Relationships: Use codes or symbols to represent relationships and ask students to decode them. 6. Puzzles with Statements: Give a set of statements about the relationships between people and have students draw a family tree or answer questions based on those statements 7. Inequality Puzzles: • Coded form of Inequalities • Either-Or Case • Neither -Nor Case Single Statement Inequalities.
3	Verbal English	Textbook: Objective General English Author: R.S Agarwal Publishing house: S. Chand	National Exams: 1. UPSC Civil Services Exam (https://upsc.gov.in/sites/default/files/Notif-CSP-23-engl- 	12	1. Application of Adjectives and Determiners: • Identification of Errors • Comparative and Superlative Forms • Types of Adjectives

		1. Application of Adjectives and Determiners 2. Conjunction and Connectors 3. Rearrangement of Sentences. 4. Multiple Fillers-Level 1 5. Reading Comprehension 6. Precise Writing Notice Writing	010223.pdf), pg 25-26 2. <i>UPSC Combined Defence Services</i> (https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf), pg 20-21 3. <i>Combined Graduate Level conducted by SSC</i> (https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_0304_2023.pdf) pg. 20-22 4. <i>Intelligence Bureau ACIO</i> (https://www.pw.live/exams/wp-content/uploads/2023/11/IB-ACIO-Recruitment-2023-Notification-Emp-News.pdf) State Level Exams: 1. <i>Civil Services Executive Exam (WBCS)</i> (https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement, pg 1 Miscellaneous Services Recruitment Examination (https://adda247jobs-wp-assets-prod.adda247.com/jobs/wp-content/uploads/sites/7/2022/11/21142422/2707970_2019.pdf) pg 1		- Determiners in Context 2. Conjunctions and Connectors: - Sentence Combining - Coordinating vs. Subordinating Conjunctions - Transition Words and Phrases - Connectors for Cause and Effect 3. Rearrangement of Sentences: - Jumbled Sentences - Paragraph Sequencing 4. Multiple Fillers - Level 1: - Cloze Passages - Sentence Completion 5. Reading Comprehension: - Inference Questions - Vocabulary in Context - Main Idea and Supporting Details - Critical Thinking Questions 6. Precise Writing: - Summarizing - Paraphrasing 8. Editing for Conciseness
4	Data Interpretation	Textbook: Quantitative Aptitude for Competitive Examination	National Exams: 1. <i>UPSC Civil Services Exam</i> (https://upsc.gov.in/sites/default	12	7. Application of Data Analysis based on Bar Chart

		Author: R.S Agarwal Publishing House: S. Chand Advanced Level: Bar Graph	<u>ult/files/Notif-CSP-23-engl-010223.pdf</u>), pg 25-26 2. <i>UPSC Combined Defence Services</i> (<u>https://upsc.gov.in/sites/default/files/Notif-CDS-I-Exam-2023-Engl-211222.pdf</u>), pg 20-21 3. <i>Combined Graduate Level conducted by SSC</i> (<u>https://ssc.nic.in/SSCFileServer/PortalManagement/UploadedFiles/notice_CGLE_03042023.pdf</u>) pg. 20-22 1. <i>Intelligence Bureau ACIO</i> (<u>https://www.pw.live/exams/wp-content/uploads/2023/11/1B-ACIO-Recruitment-2023-Notification-Emp-News.pdf</u>) 2. <i>RBI Grade B</i> (<u>https://rbidocs.rbi.org.in/rdocs/Content/PDEs/DADV TGRB09052023FA65E4FB1C2CF473396B4FD7E5F69CDDE.PDF</u>), pg 22-23 <i>State Level Exams:</i> 1. <i>Civil Services Executive Exam(WBCS)</i> (<u>https://wbpsc.gov.in/Download?param1=20230225142430_Syllabus.pdf&param2=advertisement</u>), pg 1		
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			2. Miscellaneous Services Recruitment Examination (https://adda247jobs-wp- assets- prod.adda247.com/jobs/wp- content/uploads/sites/7/2022/1 1/21142422/2707970_2019.pd f) <i>pg 1</i>		
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